Kernix 0.8.0 / Performance brief

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

21 workloads

88 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.8.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



Kernix 0.8.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



Kernix 0.8.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product

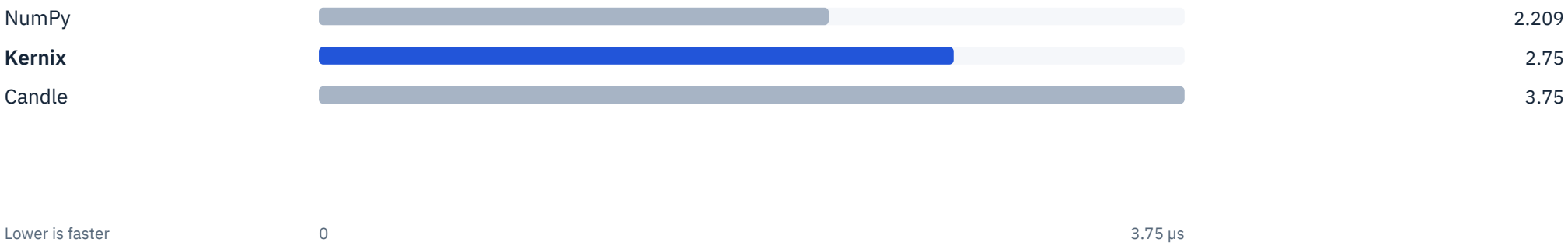


Kernix 0.8.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Kernix 0.8.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

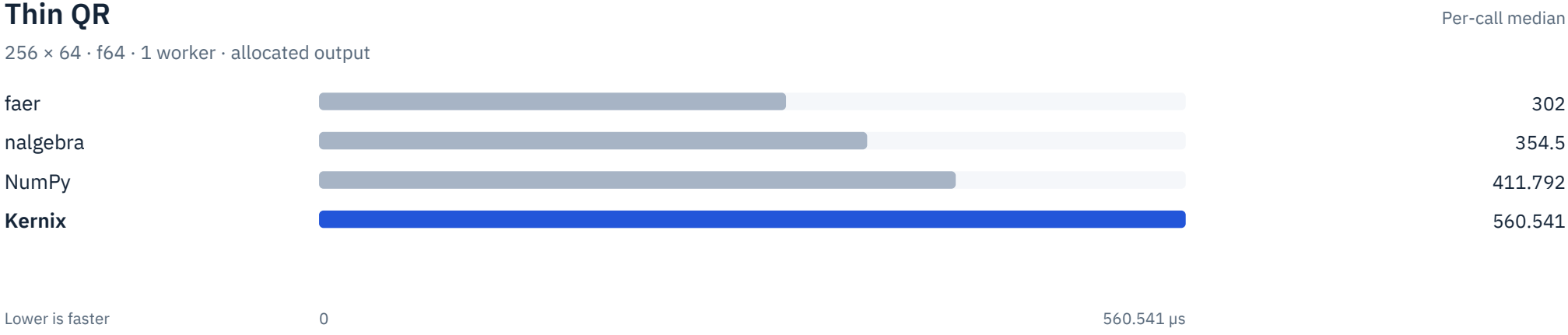
Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



Kernix 0.8.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky factorization

Per-call median

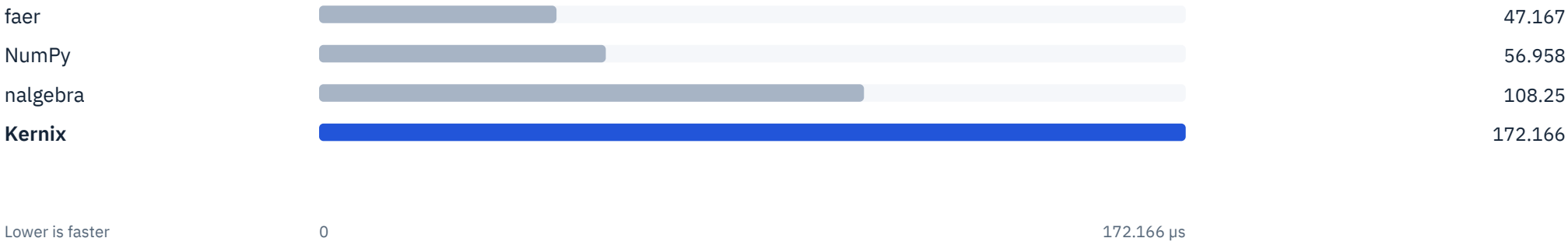
64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



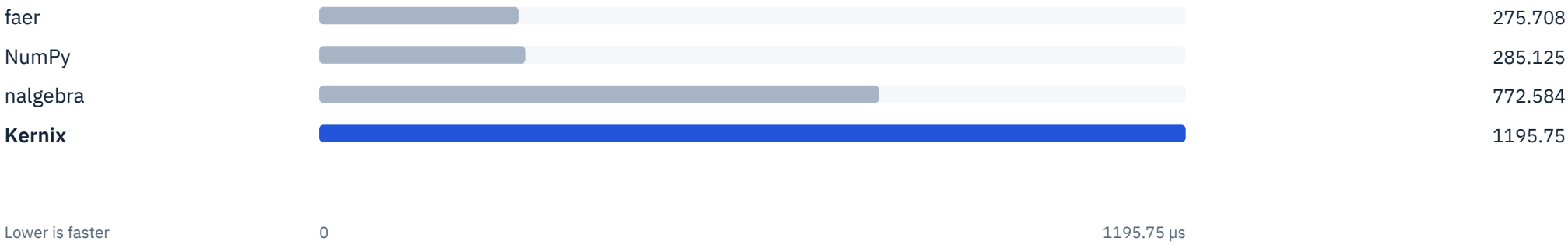
Kernix 0.8.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.8.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Record + backward

Per-call median

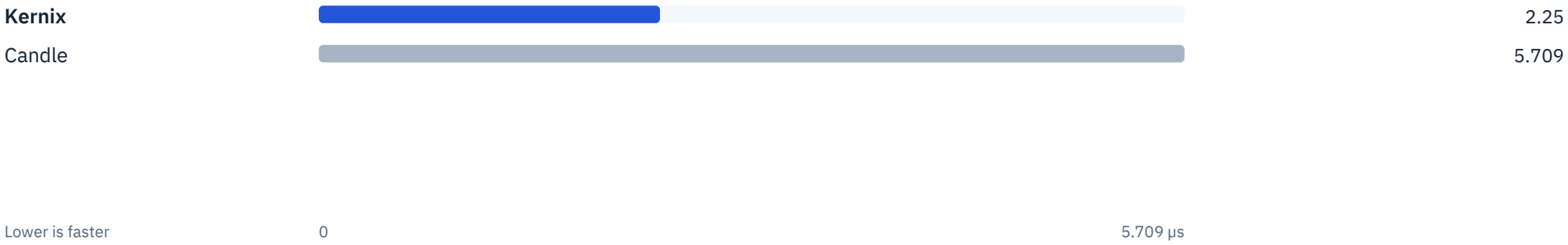
sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.8.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Model recording + forward

Per-call median

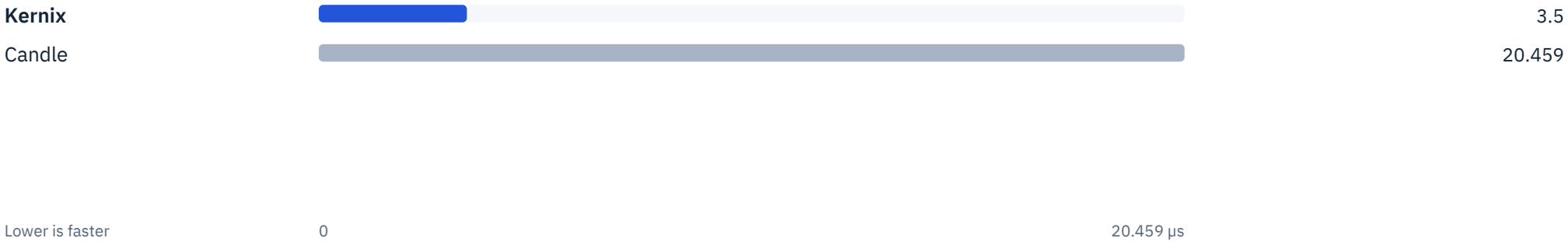
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Kernix 0.8.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Parameter update

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



Kernix 0.8.0 / Statistics & activations

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Row softmax

256 × 64 · f64 · 1 worker · allocated output

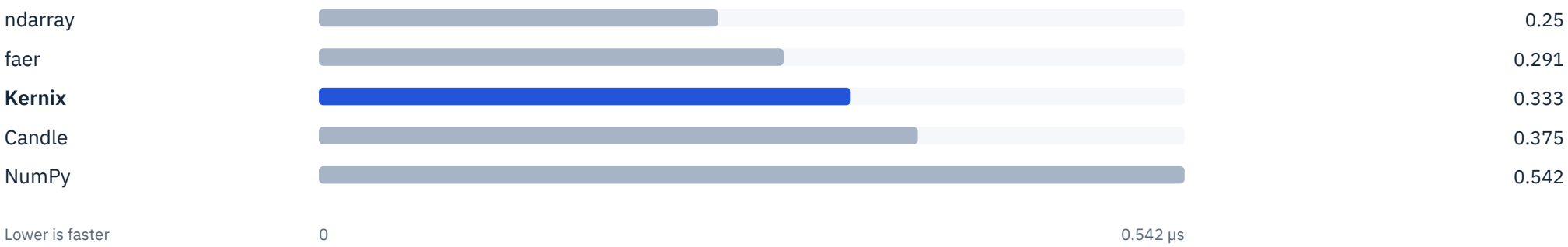


Kernix 0.8.0 / Tensor construction

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

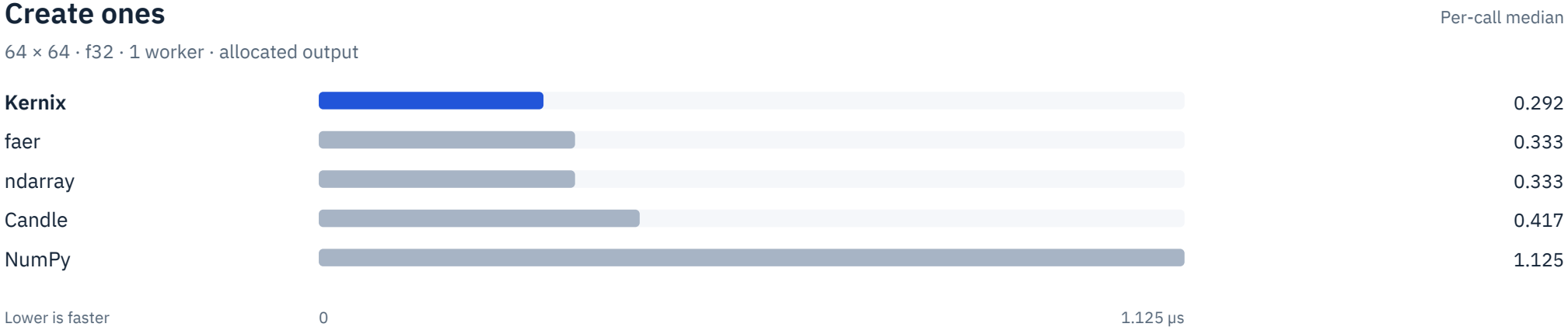
Create zeros

64 × 64 · f32 · 1 worker · allocated output



Create ones

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2250	2.25
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2041	2.041
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2417	2.417
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2542	2.542
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	1458	1.458
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2083	2.083
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	13583	13.583
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	3500	3.5
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	14083	14.083
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	13459	13.459

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	14750	14.75
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	14500	14.5
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	101292	101.292
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	104208	104.208
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	101333	101.333
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	103166	103.166
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	16125	16.125
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	97792	97.792
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	784708	784.708
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	915084	915.084

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	119833	119.833
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	818166	818.166
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	847667	847.667
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	786333	786.333
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	7084	7.084
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	7832.999999999999	7.832999999999999
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	7208	7.208
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	1750	1.75
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	8208	8.208
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	6917	6.917

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	54792	54.792
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	6083	6.083
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	51583	51.583
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	53917	53.917
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	53416	53.416
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	53333	53.333
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	470792	470.792
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	450333	450.333
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	388166	388.166
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	417625	417.625

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	34416	34.416
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	548209	548.209
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	3750	3.75
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	2209	2.209
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2750	2.75
Thin QR / Kernix 96 × 32 · f64	64708.00000000001	64.70800000000001
Thin QR / faer 96 × 32 · f64	46000	46
Thin QR / nalgebra 96 × 32 · f64	42125	42.125
Thin QR / NumPy 96 × 32 · f64	53666	53.666
Thin QR / Kernix 256 × 64 · f64	560541	560.541

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / faer 256 × 64 · f64	302000	302
Thin QR / nalgebra 256 × 64 · f64	354500	354.5
Thin QR / NumPy 256 × 64 · f64	411792	411.792
Cholesky factorization / nalgebra 64 × 64 · f64	21042	21.042
Cholesky factorization / Kernix 64 × 64 · f64	23375	23.375
Cholesky factorization / faer 64 × 64 · f64	10792	10.792
Cholesky factorization / NumPy 64 × 64 · f64	15083	15.083
Cholesky factorization / NumPy 128 × 128 · f64	56958	56.958
Cholesky factorization / nalgebra 128 × 128 · f64	108250	108.25
Cholesky factorization / faer 128 × 128 · f64	47167	47.167

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 128 × 128 · f64	172166	172.166
Cholesky factorization / faer 256 × 256 · f64	275708	275.708
Cholesky factorization / nalgebra 256 × 256 · f64	772584	772.584
Cholesky factorization / Kernix 256 × 256 · f64	1195750	1195.75
Cholesky factorization / NumPy 256 × 256 · f64	285125	285.125
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3250	3.25
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	6750	6.75
Model inference / Candle batched normalized/sliced MLP · f64	5709	5.709
Model inference / Kernix batched normalized/sliced MLP · f64	2250	2.25
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3167	3.167

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model recording + forward / Candle batched normalized/sliced MLP · f64	6208	6.208
Model backward / Candle prepared graph → fresh gradients · f64	20459	20.459
Model backward / Kernix prepared graph → fresh gradients · f64	3500	3.5
Parameter update / Kernix transactional SGD, fixed gradients · f64	709	0.709
Row softmax / NumPy 256 × 64 · f64	82458	82.458
Row softmax / Candle 256 × 64 · f64	70041	70.041
Row softmax / Kernix 256 × 64 · f64	78583	78.583
Row softmax / ndarray 256 × 64 · f64	61667.00000000001	61.66700000000001
Create zeros / Kernix 64 × 64 · f32	333	0.333
Create zeros / faer 64 × 64 · f32	291	0.291

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / ndarray 64 × 64 · f32	250	0.25
Create zeros / NumPy 64 × 64 · f32	542	0.542
Create zeros / Candle 64 × 64 · f32	375	0.375
Create ones / faer 64 × 64 · f32	333	0.333
Create ones / Kernix 64 × 64 · f32	292	0.292
Create ones / Candle 64 × 64 · f32	417	0.417
Create ones / NumPy 64 × 64 · f32	1125	1.125
Create ones / ndarray 64 × 64 · f32	333	0.333